



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

CHEMISTRY

PAPER 1 Multiple Choice

4024/1

1 hour

JUNE 2026 SESSION

Additional materials:

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

Electronic calculator BY97

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the correct answer and record your choice on the separate answer sheet.

INFORMATION FOR CANDIDATES

A copy of the Periodic Table is printed on page 16.

This question paper consists of 16 printed pages.

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1. Which apparatus is used to measure precise volumes of liquids?
 - A beaker
 - B measuring cylinder
 - C burette
 - D pipette

2. A mixture of ethanol and water can be separated using fractional distillation because they are
 - A immiscible and high temperature sensitive.
 - B miscible and high temperature sensitive.
 - C miscible and have different boiling points.
 - D immiscible and have different boiling points.

3. Which process occurs when iodine vapour is cooled?
 - A condensation
 - B deposition
 - C freezing
 - D sublimation



4. The table shows the proton numbers of two elements, X and Y.

element	proton number
X	6
Y	9

What is the chemical formula of the compound formed when X and Y chemically combine?

- A X_2Y_3
- B X_3Y_2
- C XY_3
- D XY_4
5. Which gases diffuse at the same rate at room temperature and pressure?
- A Cl_2 and F_2
- B CO and CO_2
- C CO and N_2
- D N_2 and O_2



6. The table shows the electronic configurations of four elements J, K, L and M.

element	electronic configuration
J	2.7
K	2.8.7
L	2.8.8.7
M	2.8.8.8.7

The elements represented by the letters J, K, L and M belong to the Group of

- A alkaline earth metals.
 - B alkali metals.
 - C halogens.
 - D noble gases.
7. Gallium, *Ga*, forms an oxide of molecular formula Ga_2O_3 .
The bonding in Ga_2O_3 is

- A covalent.
 - B ionic.
 - C dative.
 - D metallic.
8. How many atoms of silicon are there in 25 g of silicon ?

- A 6.02×10^{23}
- B 5.38×10^{23}
- C 6.74×10^{23}
- D 3.37×10^{23}



9. Which substance has the smallest mass?
- A 0.1 moles of SiO_2
 - B 0.5 moles of O_2
 - C 0.5 moles of Li
 - D 1.0 moles of NH_3
10. Which metal exist as a free element in the earth's crust?
- A aluminium
 - B nickel
 - C lithium
 - D gold
11. The relative molecular mass, M_r , of hydrated copper (II) nitrate, $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ is
- A 16.
 - B 144.
 - C 198.
 - D 242.
12. A mass of 2.00 g of sodium hydroxide, NaOH , was dissolved in 250 cm^3 of distilled water.
- The concentration of the resultant solution in mol/dm^3 was
- A 0.2000.
 - B 0.0200.
 - C 0.0020.
 - D 0.0002.

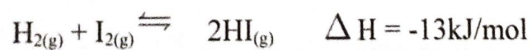


13. Ethene is converted to ethane according to the equation:



What is the volume of hydrogen required to react with 14.00 g of ethene at *r.t.p*?

- A 1.400 dm³
B 14.00 dm³
C 0.014 dm³
D 0.140 dm³
14. A mass of 6.50 g of ethanol was used to heat 250.00 cm³ of water from 22.50 °C to 35.00 °C.
How much energy was used up?
- A 13 125 J
B 13 466 J
C 341 J
D 13 J
15. Hydrogen reacts with iodine according to the equation shown.

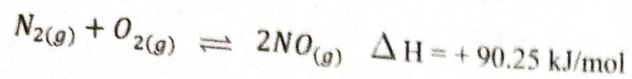


Which action gives a higher yield of *HI*?

- A decreasing the pressure
B decreasing the temperature
C increasing the pressure
D increasing the temperature



16. The equation shows a reversible reaction:



Which option correctly describes the effect of increasing temperature in the reaction?

- A favours forward reaction
 - B favours backward reaction
 - C yield of NO decreases
 - D rate of formation of NO decreases
17. What is the use of calcium oxide in steel making?
- A neutralises silicon dioxide
 - B oxidises silicon dioxide
 - C reduces silicon dioxide
 - D dissolves silicon dioxide
18. Which equation is correct, for the removal of impurities in the blast furnace?
- A $CaO_{(s)} + SiO_{2(s)} \longrightarrow CaSiO_{3(l)}$
 - B $CO_{2(g)} + C_{(s)} \longrightarrow 2CO_{(s)}$
 - C $CaCO_{(s)} \longrightarrow CaO_{(s)} + CO_{2(g)}$
 - D $Fe_2O_{3(s)} + 3CO_{(g)} \longrightarrow 2Fe_{(s)} + 3CO_{2(g)}$
19. Which option correctly indicates decreasing reactivity?

- A Ca, Zn, Fe, Ag
- B Zn, Ca, Ag, Fe
- C Ca, Ag, Zn, Fe
- D Ag, Ca, Fe, Zn



20. A metal, **M**, reacts with atmospheric oxygen to form an oxide layer that prevents the metal from being reactive.

Metal **M** is

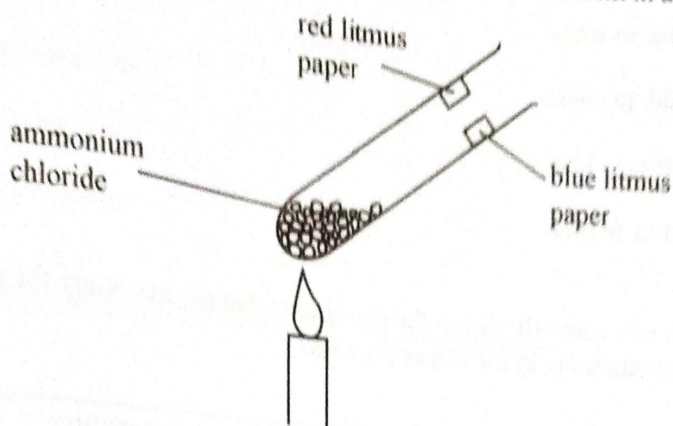
- A iron.
- B aluminium.
- C copper.
- D silver.

21. Which one is a product of the reaction between AgNO_3 and Zn ?

- A $\text{Ag}^+_{(\text{aq})}$
- B $\text{H}_{2(\text{g})}$
- C $\text{Ag}_{(\text{s})}$
- D $\text{Zn}_{(\text{s})}$



22. The diagram shows a set-up of an experiment carried out by a student in a laboratory.



Which chemical change takes place?

- A $\text{NH}_4\text{Cl}_{(\text{aq})} \longrightarrow \text{NH}_4^+_{(\text{aq})} + \text{Cl}^-$
- B $\text{NH}_4^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})} \longrightarrow \text{NH}_4\text{Cl}_{(\text{aq})}$
- C $\text{NH}_4\text{Cl}_{(\text{s})} \longrightarrow \text{NH}_{3(\text{g})} + \text{HCl}_{(\text{g})}$
- D $\text{NH}_4\text{Cl}_{(\text{g})} \longrightarrow \text{NH}_4\text{Cl}_{(\text{s})}$
23. Which fuel is obtained by destructive distillation of coal?
- A charcoal
- B biogas
- C coke
- D ethanol
24. The energy profile diagram for an endothermic reaction shows that the
- A products are at a lower energy level.
- B products are at a higher energy level.
- C reactants and products are at the same energy level.
- D activation energy is zero.



25. What colour change is observed when water is added to anhydrous copper (II) sulphate?

- A blue to white
- B pink to white
- C white to blue
- D white to pink

26. Which row correctly shows the conditions that are necessary for achieving an economic yield of ammonia in the Haber process?

	pressure	temperature
A	high	high
B	low	high
C	high	low
D	low	low

27. Aluminium **cannot** be extracted from its ores by reduction because aluminium is

- A below carbon in the reactivity series.
- B above carbon in the reactivity series.
- C more reactive than carbon.
- D less reactive than carbon.

28. Which option, A, B, C or D correctly shows the conditions for the manufacture of nitric acid?

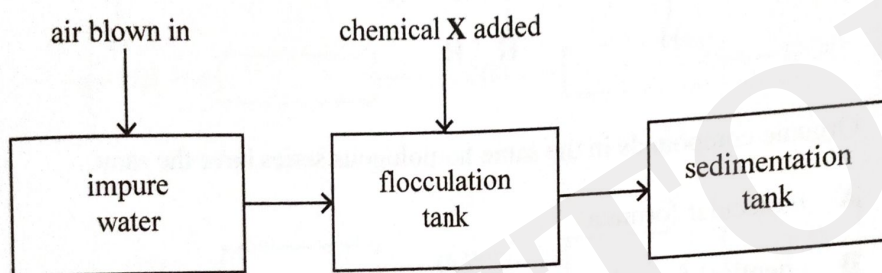
	pressure/atm	temperature/°C	catalyst
A	2	450	vanadium oxide
B	200	450	iron oxide
C	4-6	90	platinum/rhodium
D	4-6	900	platinum/rhodium



29. Saponification is an example of

- A a hydrolysis reaction.
- B an oxidation reaction.
- C a hydration reaction.
- D an addition reaction.

30. The flow diagram represents some stages in water purification.

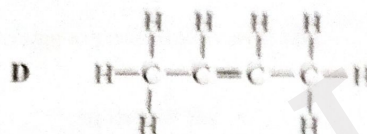
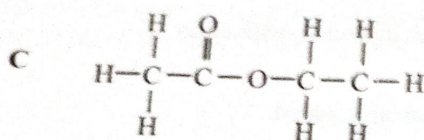
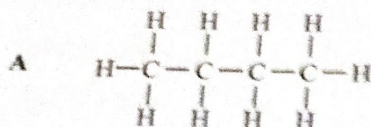


Chemical X is

- A aluminium nitrate.
- B aluminium sulphate.
- C chlorine.
- D carbon.



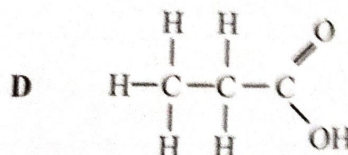
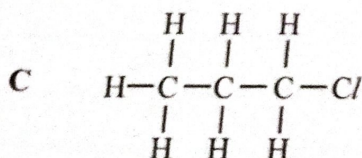
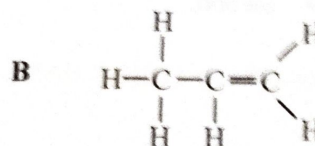
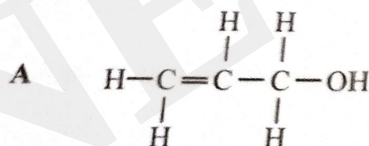
31. Which compound can be oxidised to a carboxylic acid?



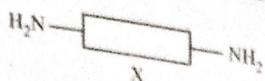
32. Organic compounds in the same homologous series have the same

- A molecular formula
- B chemical formula
- C number of isomers
- D empirical formula

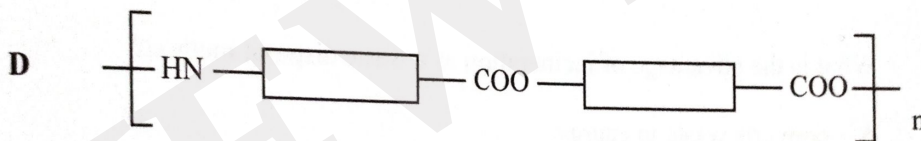
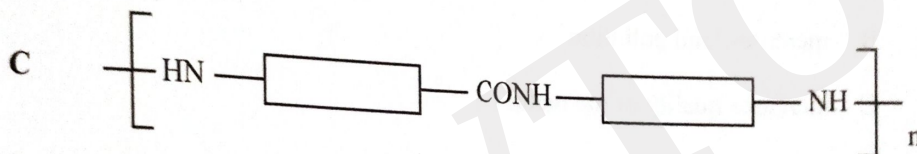
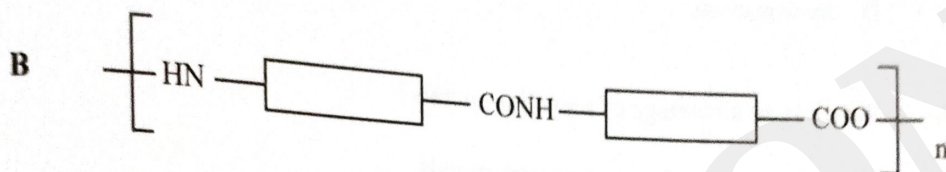
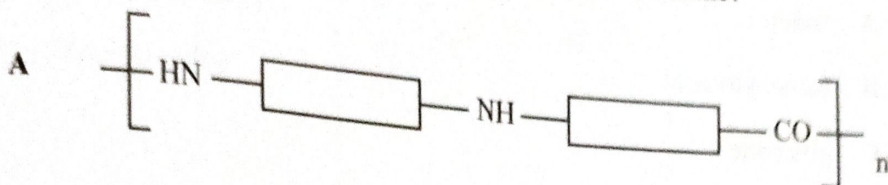
33. Which compound is a hydrocarbon?



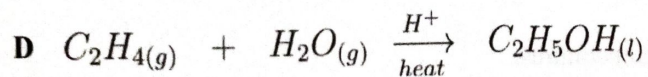
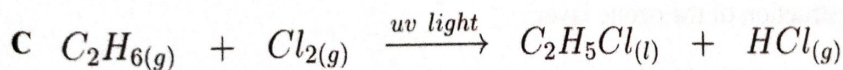
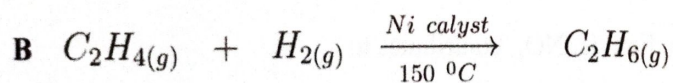
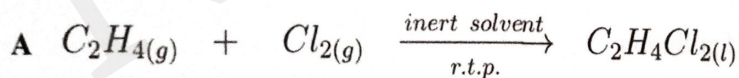
34. The diagram shows the structural formula of two monomers X and Y.



What is the structure of a polymer formed when X and Y combine?



35. Which reaction is a substitution reaction?



36. Biodiesel is produced by chemically reacting a vegetable oil or animal fat with an alcohol.
- Which other product is produced from the reaction?
- A water
 - B carboxylic acid
 - C glycerine
 - D hydrocarbon
37. Which is an advantage of recycling metals?
- A reduces cost of extracting new metals
 - B increases land pollution
 - C increases quality of the metal
 - D reduces costs of the products
38. What is the advantage of incineration as a waste disposal method?
- A converts waste to energy
 - B produces gaseous pollutants
 - C produces ash
 - D reduces land pollution
39. The pollutant nitrogen dioxide, NO_2 , contributes to
- A destruction of the ozone layer.
 - B formation of acid rain.
 - C increase in global temperatures.
 - D asthma attacks.



40. Which pollutant is **not** a product of the burning fuel in automobiles?
- A carbon monoxide
 - B hydrocarbons
 - C lead compounds
 - D nitrogen oxides

NEWTON



DATA SHEET

The Periodic Table of the Elements



18324



Zimsec 2026

Group

I		II		Group										III	IV	V	VI	VII	O																											
7 Li Lithium 3 9 Be Beryllium 4		11 Na Sodium 23 12 Mg Magnesium 24		1 H Hydrogen 1										13 Al Aluminium 27	14 Si Silicon 28	15 P Phosphorus 31	16 S Sulphur 32	17 Cl Chlorine 35.5	18 Ar Argon 40	19 K Potassium 39 20 Ca Calcium 40	21 Sc Scandium 45 22 Ti Titanium 48	23 V Vanadium 51 24 Cr Chromium 52	25 Mn Manganese 55 26 Fe Iron 56	27 Co Cobalt 59 28 Ni Nickel 59	29 Cu Copper 64 30 Zn Zinc 65	31 Ga Gallium 70 32 Ge Germanium 73	33 As Arsenic 75 34 Se Selenium 79	35 Br Bromine 80 36 Kr Krypton 84	37 Rb Rubidium 85 38 Sr Strontium 88	39 Y Yttrium 89 40 Zr Zirconium 91	41 Nb Niobium 93 42 Mo Molybdenum 96	43 Tc Technetium 98 44 Ru Ruthenium 101	45 Rh Rhodium 103 46 Pd Palladium 106	47 Ag Silver 108 48 Cd Cadmium 112	49 In Indium 115 50 Sn Tin 119	51 Sb Antimony 122 52 Te Tellurium 128	53 I Iodine 127 54 Xe Xenon 131	55 Cs Caesium 132 56 Ba Barium 137	57 La Lanthanum 139 58 Ce Cerium 140	59 Pr Praseodymium 141 60 Nd Neodymium 144	61 Pm Promethium 147 62 Sm Samarium 150	63 Eu Europium 152 64 Gd Gadolinium 157	65 Tb Terbium 159 66 Dy Dysprosium 162	67 Ho Holmium 165 68 Er Erbium 167	69 Tm Thulium 169 70 Yb Ytterbium 173	71 Lu Lutetium 175
Fr Francium 87		Ra Radium 88		Ac Actinium 89																																										
58-71 Lanthanoid series		90-103 Actinoid series																																												
Key		X = relative atomic mass																																												
X = atomic symbol		p = proton (atomic) Number																																												
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